

TP 5: Apache Spark

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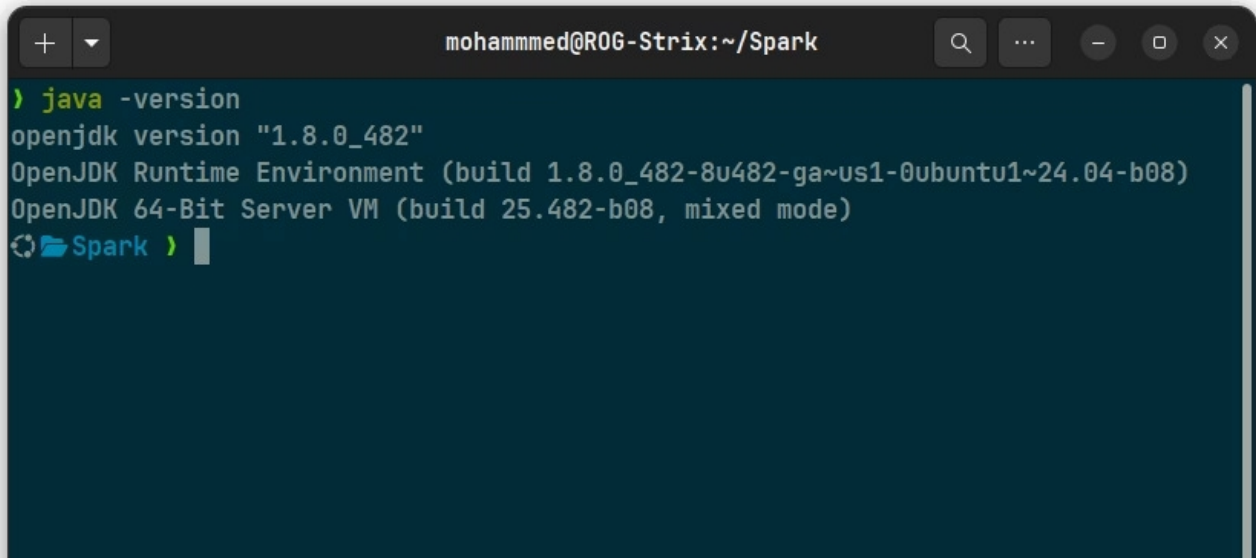
- 2025–2026

Installation de Apache Spark

Étape 1 : téléchargement

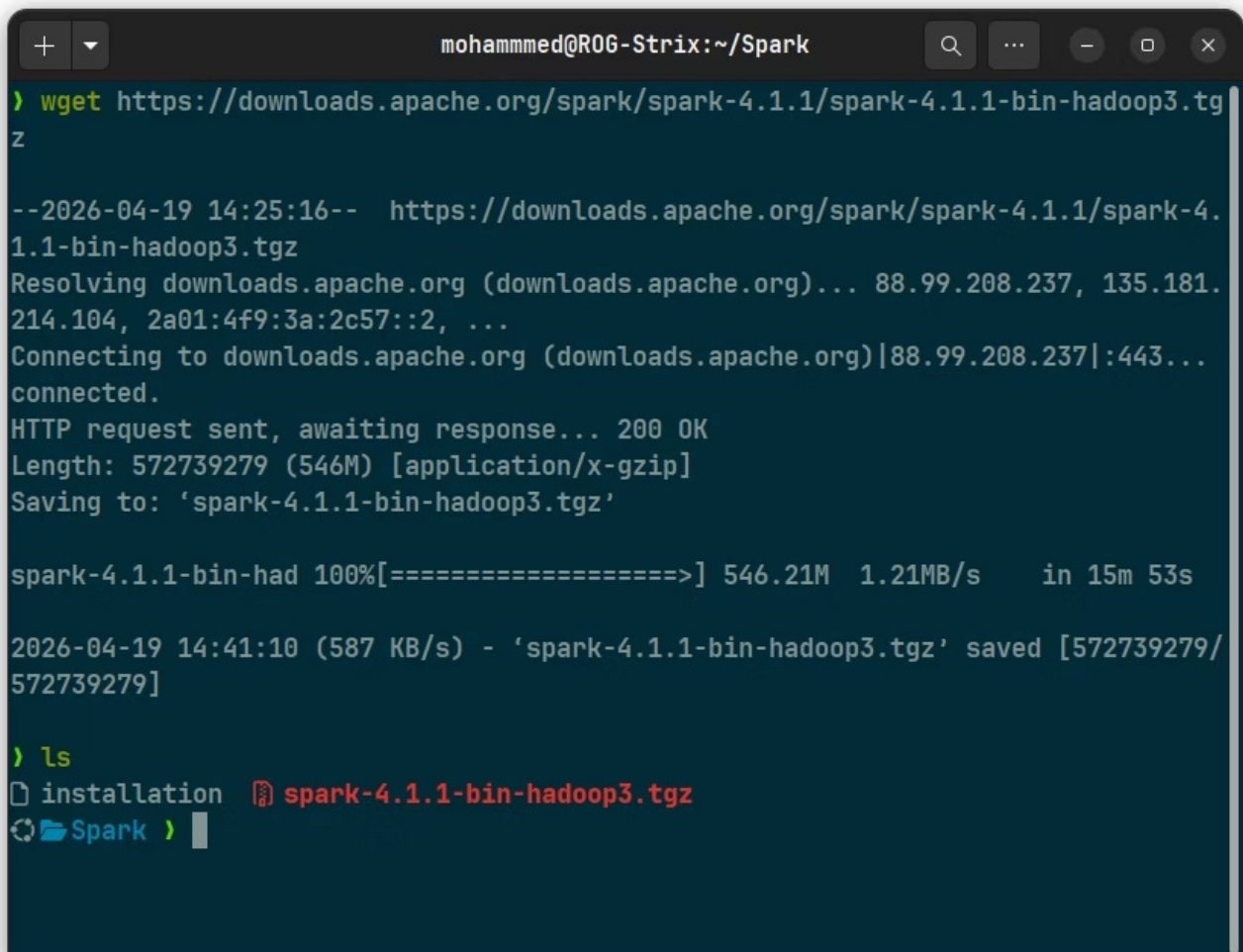
1. Installing Java(JDK)

- java is already installed

A terminal window with a dark blue background. The title bar shows 'mohammed@ROG-Strix:~/Spark'. The command 'java -version' has been executed, resulting in the following output: 'openjdk version "1.8.0_482"', 'OpenJDK Runtime Environment (build 1.8.0_482-8u482-ga~us1-0ubuntu1~24.04-b08)', and 'OpenJDK 64-Bit Server VM (build 25.482-b08, mixed mode)'. The prompt is now 'Spark >'.

```
mohammed@ROG-Strix:~/Spark
> java -version
openjdk version "1.8.0_482"
OpenJDK Runtime Environment (build 1.8.0_482-8u482-ga~us1-0ubuntu1~24.04-b08)
OpenJDK 64-Bit Server VM (build 25.482-b08, mixed mode)
Spark >
```

2. installing Apache Spark

A terminal window with a dark blue background. The title bar shows 'mohammed@ROG-Strix:~/Spark'. The command 'wget https://downloads.apache.org/spark/spark-4.1.1/spark-4.1.1-bin-hadoop3.tgz' has been executed. The output shows the download progress, including the URL, file size (546M), and speed (1.21MB/s). The file is saved as 'spark-4.1.1-bin-hadoop3.tgz'. The prompt is now 'Spark >'.

```
mohammed@ROG-Strix:~/Spark
> wget https://downloads.apache.org/spark/spark-4.1.1/spark-4.1.1-bin-hadoop3.tgz
--2026-04-19 14:25:16-- https://downloads.apache.org/spark/spark-4.1.1/spark-4.1.1-bin-hadoop3.tgz
Resolving downloads.apache.org (downloads.apache.org)... 88.99.208.237, 135.181.214.104, 2a01:4f9:3a:2c57::2, ...
Connecting to downloads.apache.org (downloads.apache.org)|88.99.208.237|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 572739279 (546M) [application/x-gzip]
Saving to: 'spark-4.1.1-bin-hadoop3.tgz'

spark-4.1.1-bin-had 100%[=====>] 546.21M  1.21MB/s   in 15m 53s

2026-04-19 14:41:10 (587 KB/s) - 'spark-4.1.1-bin-hadoop3.tgz' saved [572739279/572739279]

> ls
installation  spark-4.1.1-bin-hadoop3.tgz
Spark >
```

Installation de Apache Spark

Étape 2: Configuration des Variables d'Environnement

1. Extract the file

```
mohammed@ROG-Strix:~/Spark
> tar -xvf spark-4.1.1-bin-hadoop3.tgz
spark-4.1.1-bin-hadoop3/
spark-4.1.1-bin-hadoop3/conf/
spark-4.1.1-bin-hadoop3/conf/workers.template
spark-4.1.1-bin-hadoop3/conf/fairscheduler.xml.template
spark-4.1.1-bin-hadoop3/conf/spark-defaults.conf.template
spark-4.1.1-bin-hadoop3/conf/log4j2.properties.template
spark-4.1.1-bin-hadoop3/conf/log4j2-json-layout.properties.template
spark-4.1.1-bin-hadoop3/conf/spark-env.sh.template
spark-4.1.1-bin-hadoop3/conf/metrics.properties.template
spark-4.1.1-bin-hadoop3/NOTICE
spark-4.1.1-bin-hadoop3/jars/
spark-4.1.1-bin-hadoop3/jars/datasketches-memory-3.0.2.jar
spark-4.1.1-bin-hadoop3/jars/jline-2.14.6.jar
spark-4.1.1-bin-hadoop3/jars/kubernetes-model-autoscaling-7.4.0.jar
spark-4.1.1-bin-hadoop3/jars/javassist-3.30.2-GA.jar
spark-4.1.1-bin-hadoop3/jars/spark-network-shuffle_2.13-4.1.1.jar
spark-4.1.1-bin-hadoop3/jars/netty-handler-4.2.7.Final.jar
spark-4.1.1-bin-hadoop3/jars/orc-shims-2.2.1.jar
spark-4.1.1-bin-hadoop3/jars/hadoop-client-api-3.4.2.jar
spark-4.1.1-bin-hadoop3/jars/spark-yarn_2.13-4.1.1.jar
spark-4.1.1-bin-hadoop3/jars/avro-1.12.1.jar
spark-4.1.1-bin-hadoop3/jars/spark-kvstore_2.13-4.1.1.jar
spark-4.1.1-bin-hadoop3/jars/jakarta.servlet-api-5.0.0.jar
```

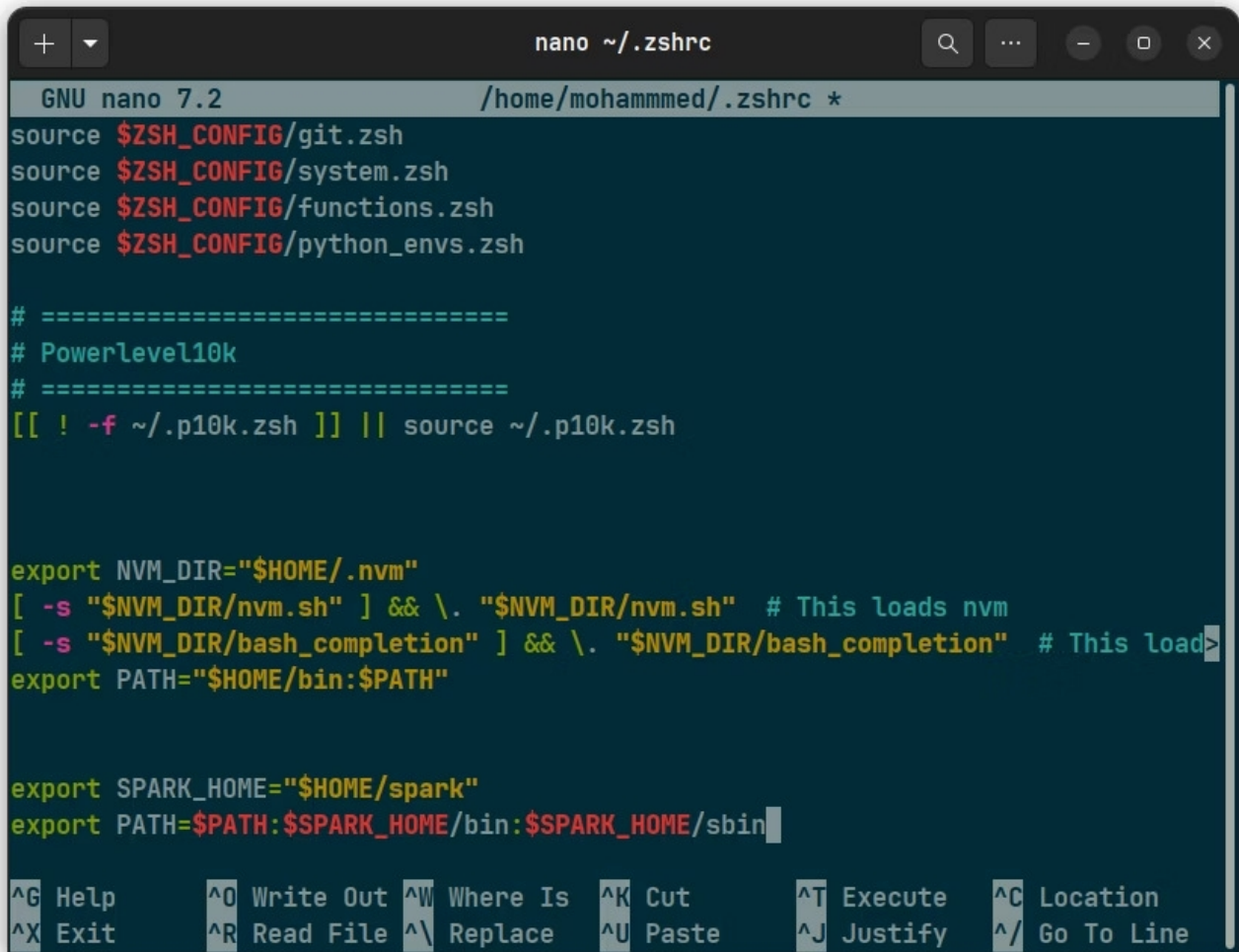
2. Move Spark to ~/

```
mohammed@ROG-Strix:~/Spark
> mv spark-4.1.1-bin-hadoop3 ~/spark
🔄 Spark >
```

Installation de Apache Spark

Étape 3: Configuration des Variables d'Environnement

3. Set environment variables

A screenshot of a terminal window with the nano text editor open, editing the file ~/.zshrc. The editor's title bar shows 'nano ~/.zshrc'. The content of the file includes source commands for \$ZSH_CONFIG sub-files, a Powerlevel10k configuration block, and environment variable exports for NVM, SPARK_HOME, and PATH. The SPARK_HOME is set to \$HOME/spark and its bin and sbin directories are added to the PATH. A help footer is visible at the bottom of the editor.

```
GNU nano 7.2 /home/mohammed/.zshrc *
source $ZSH_CONFIG/git.zsh
source $ZSH_CONFIG/system.zsh
source $ZSH_CONFIG/functions.zsh
source $ZSH_CONFIG/python_envs.zsh

# =====
# Powerlevel10k
# =====
[[ ! -f ~/.p10k.zsh ]] || source ~/.p10k.zsh

export NVM_DIR="$HOME/.nvm"
[ -s "$NVM_DIR/nvm.sh" ] && \. "$NVM_DIR/nvm.sh" # This loads nvm
[ -s "$NVM_DIR/bash_completion" ] && \. "$NVM_DIR/bash_completion" # This load
export PATH="$HOME/bin:$PATH"

export SPARK_HOME="$HOME/spark"
export PATH=$PATH:$SPARK_HOME/bin:$SPARK_HOME/sbin

^G Help      ^O Write Out ^W Where Is  ^K Cut       ^T Execute  ^C Location
^X Exit      ^R Read File ^\ Replace   ^U Paste     ^J Justify  ^_ Go To Line
```

3. Apply changes

A screenshot of a terminal window showing the command 'source ~/.zshrc' being executed. The terminal title bar shows 'mohammed@ROG-Strix:~/Spark'. Below the command, there is a prompt 'Spark >' with a cursor.

```
mohammed@ROG-Strix:~/Spark
> source ~/.zshrc
Spark > 
```

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Étape 2: Vérification et Lancement

```
spark-shell
```

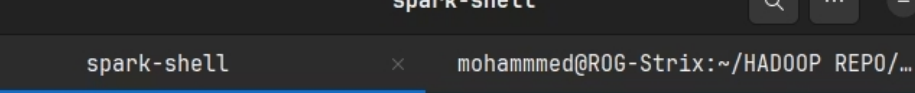
```
26/04/19 15:59:12 WARN Utils: Set SPARK_LOCAL_IP if you need to bind to another address  
Using Spark's default log4j profile: org/apache/spark/log4j2-defaults.properties  
Setting default log level to "WARN".  
To adjust logging level use sc.setLogLevel(newLevel). For SparkR, use setLogLevel(newLevel).  
Welcome to
```

```
      _--_          _--_  
     /  _ \__    __/  _ \__    __/  
    _\ V _ \_  _/_\ V _ \_  _/  
   /___/\___\/___\/___\/___\/___\ version 4.1.1  
      _/_
```

```
Using Scala version 2.13.17 (OpenJDK 64-Bit Server VM, Java 17.0.18)  
Type in expressions to have them evaluated.  
Type :help for more information.  
26/04/19 15:59:15 WARN NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-java classes where applicable  
Spark context Web UI available at http://192.168.1.7:4040  
Spark context available as 'sc' (master = local[*], app id = local-1776610756318).  
Spark session available as 'spark'.  
  
scala>
```

- Vérification :

```
Println("Spark is working")
```

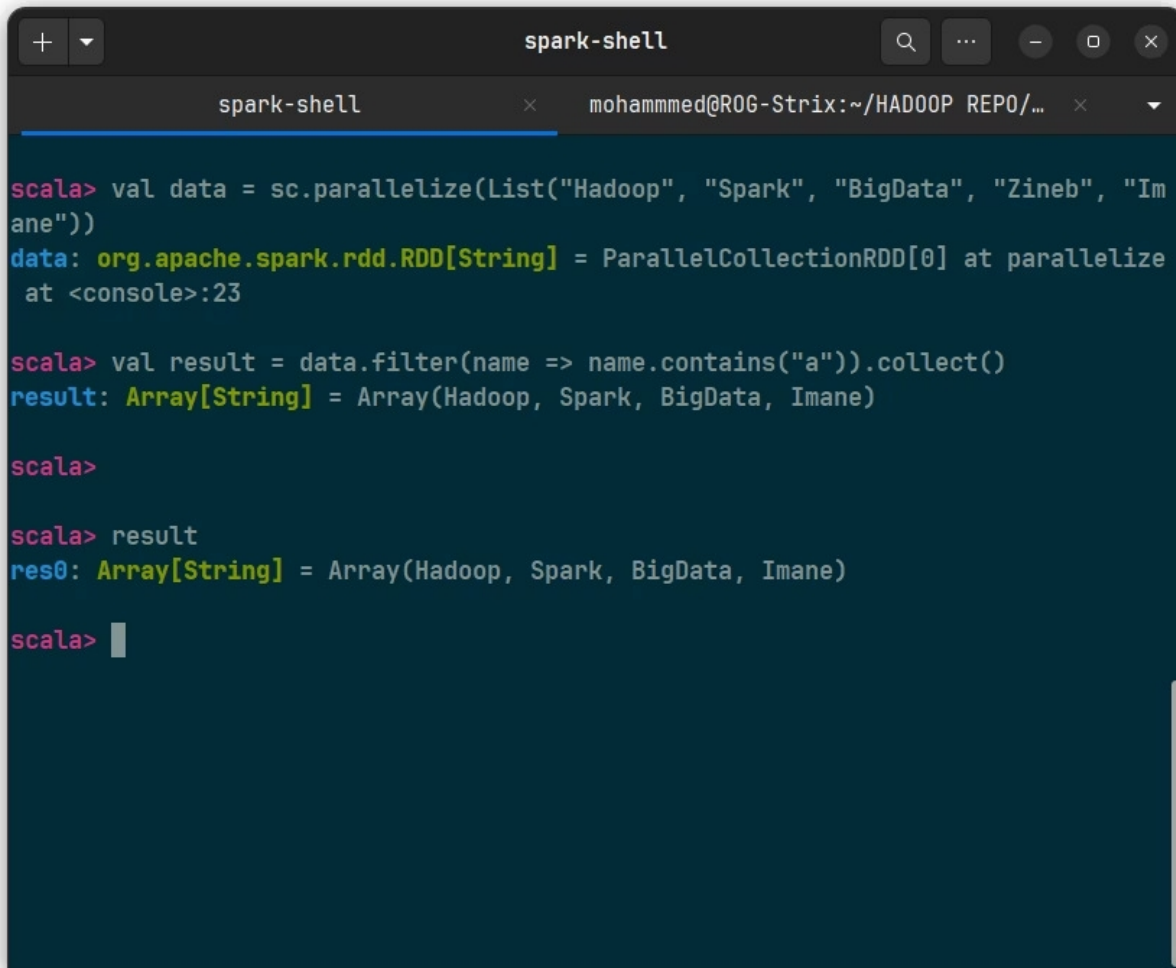


The screenshot shows a terminal window titled "spark-shell". The window has a dark theme. The terminal content shows a Scala prompt "scala>" followed by the command "println(" Spark is working")". The output "Spark is working" is displayed on the next line. The prompt "scala>" is shown again on the third line, followed by a vertical bar cursor.

```
spark-shell
scala> println(" Spark is working")
Spark is working
scala> |
```


Manipulation des packages

1. filtrer ceux qui contiennent la lettre "a"



```
spark-shell
scala> val data = sc.parallelize(List("Hadoop", "Spark", "BigData", "Zineb", "Imane"))
data: org.apache.spark.rdd.RDD[String] = ParallelCollectionRDD[0] at parallelize at <console>:23

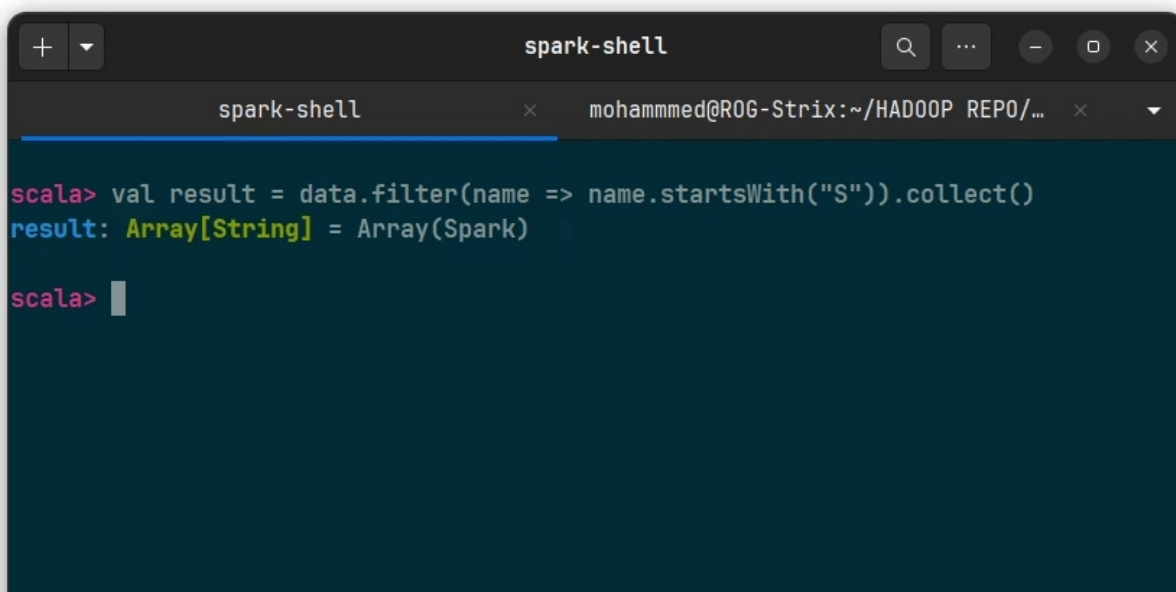
scala> val result = data.filter(name => name.contains("a")).collect()
result: Array[String] = Array(Hadoop, Spark, BigData, Imane)

scala>

scala> result
res0: Array[String] = Array(Hadoop, Spark, BigData, Imane)

scala>
```

2. Trouver les mots qui commencent par "S"



```
spark-shell
scala> val result = data.filter(name => name.startsWith("S")).collect()
result: Array[String] = Array(Spark)

scala>
```